Credit assignment, five ways, two environments

token-PPO, turn-PPO, GRPO, GiGPO and HCAPO on SearchQA and ALFWorld: matched-budget training curves and final evaluations

Agentic-RL-CA project

2026-08-14 (draft; final after the two in-flight SearchQA arms complete)

Abstract

We compare five credit-assignment strategies for multi-turn agentic RL — token-level PPO (GAE over tokens, learned critic), turn-level PPO (GAE over turns, learned critic), GRPO (trajectory-level group baseline), GiGPO (episode-level group baseline plus anchor-state step-level groups) and HCAPO (hindsight credit reweighting) — on two environments with per-environment matched budgets and configurations. On **ALFWorld** (Qwen3-1.7B from a shared replay-BC initialisation, unseen 134-game evaluation) the ordering is turn-PPO 0.963 > GiGPO 0.955 > GRPO 0.881 > HCAPO 0.784 > token-PPO 0.761. On **SearchQA** (Qwen3-4B, 4-turn non-thinking protocol, 500 steps, full 51,713-question evaluation) the completed arms give GRPO 0.463 > HCAPO(adapted) 0.440 > turn-PPO 0.417 > HCAPO(paper) 0.410; *token-PPO and GiGPO are training now (launched 2026-08-14, config-matched) and will complete this table.* Two observations already stand: **(1) no single credit scheme wins both environments** — turn-level PPO dominates the long-horizon, sparse-reward ALFWorld but is mid-pack on short-horizon SearchQA, where the critic-free group baseline (GRPO) leads; **(2) HCAPO trails flat GRPO in both environments** under matched budgets, consistent with the dedicated HCAPO analysis (2026-08-03 report).

1 Setup

1.1 Environments and protocols

SearchQA (Search-R1 style): the agent interleaves `<search>` calls against a wiki-18 e5 retriever (top-3) with a final `<answer>`; max 4 turns; Qwen3-4B, non-thinking chat template, 2,048-token response cap; 500 training steps at 1,280 trajectories/step (batch 256 × group 5), lr 10^{-6}, KL 10^{-3}; validation every 25 steps is greedy strict-EM on a fixed 2,048-question slice; the final evaluation is the full 51,713-question 7-task suite (NQ, TriviaQA, PopQA, HotpotQA, 2WikiMultiHopQA, Musique, Bamboogle), reported as macro EM over tasks. All five arms share the protocol file (`configs/protocol_4turn_nothink2k.sh`) and differ only in `algorithm.adv_estimator` (plus the critic for the two PPO arms).

ALFWorld: embodied household tasks, ReAct full-transcript prompt, max 50 turns; Qwen3-1.7B initialised from a shared replay-BC checkpoint (unseen success 0.440); 16 trajectory-groups of 16 (PPO arms: 128 envs), lr 10^{-6}, KL-to-π_{base} 0.01; validation every 5 steps on the seen split; the final evaluation is greedy success on the full 134-game out-of-distribution split at each method's best checkpoint.

1.2 Methods and caveats

The five estimators map to `gae` (token-PPO), `gae_turn` (turn-PPO), `token_grpo` (GRPO), `gigpo` (GiGPO, mean-std normalisation, step-advantage weight 1.0) and `hcapo`. HCAPO appears twice where two variants exist: the paper-faithful configuration (*HCAPO*) and the environment-adapted variant that scored best in the earlier per-environment studies (*HCAPO (adapted)*: hindsight-source `answer` on SearchQA; the ω=1, T=5 clip variant on ALFWorld).

Caveats, stated once and applying throughout:

- The two environments use *different model scales* (4B vs 1.7B) and eval metrics; compare orderings within an environment, never numbers across environments.
- Single seed per arm. The SearchQA greedy-val reproducibility band is ±0.02; deltas inside it are noise.
- ALFWorld training curves plot the *seen*-split validation; the table reports the *unseen* 134-game split, which is the cross-method-comparable number.
- The SearchQA token-PPO and GiGPO cells were trained in this campaign (2026-08-14, ark H100 queue, checkpoint-resume across 4-hour pod reclamations) with configurations matched 1:1 to the July non-thinking arms; all other cells reuse the completed July–August runs unchanged.

2 SearchQA

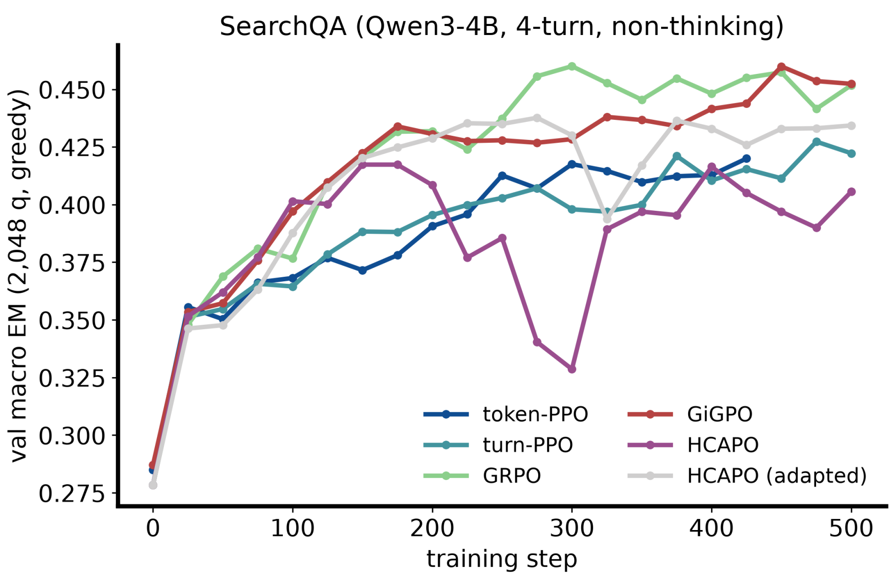

Figure 1: **SearchQA, Qwen3-4B, 4-turn non-thinking, 500 steps.** Greedy validation macro-EM (2,048 questions) every 25 steps. *The token-PPO curve completes when its arm finishes.*

method	macro EM	eval step
turn-PPO	0.417	500
GRPO	**0.463**	500
GiGPO	0.454	500
HCAPO	0.410	500
HCAPO (adapted)	0.440	500
pre-RL init	0.313	0

Table 1: **SearchQA final evaluation:** full-set (51,713-question) macro EM at step 500. Best non-baseline value in bold, second best underlined. *token-PPO row appears when its arm completes.*

Among completed arms, GRPO leads at 0.463 macro EM, +0.046 over turn-PPO and +0.023 over the best HCAPO variant. The HCAPO curves show the documented response-length runaway dips (paper variant dips hardest around step 300) yet recover to a plateau below GRPO. *Token-PPO and GiGPO commentary pending run completion; the thinking-protocol precedent (GiGPO* 0.446 *vs GRPO* 0.441*, token-PPO* 0.381*) suggests GiGPO lands near GRPO and token-PPO lands near the bottom of the table.*

3 ALFWorld

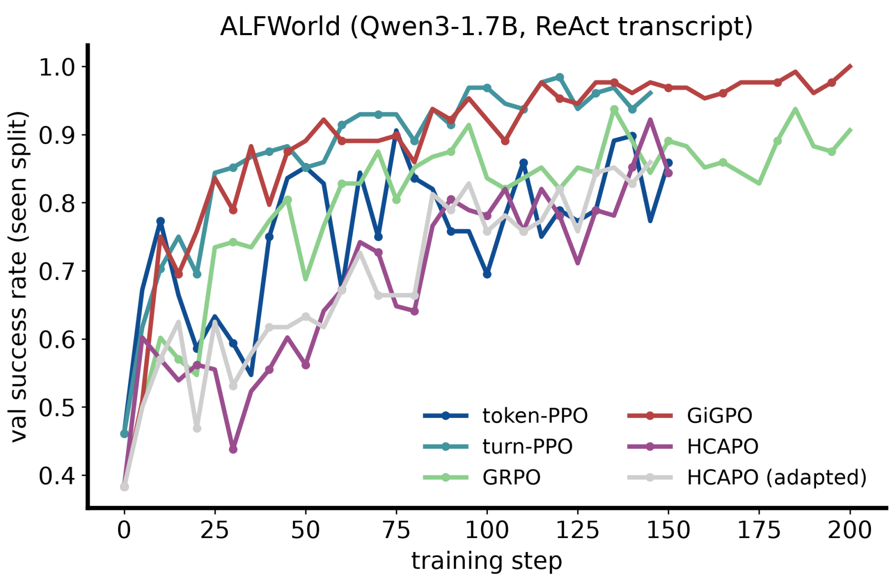

Figure 2: **ALFWorld, Qwen3-1.7B from replay-BC.** Seen-split validation success (logged every 5 steps; curves truncated at step 200 for readability — PPO/HCAPO arms trained 150 steps, GRPO/GiGPO 300, budgets follow the original campaign).

method	unseen success	eval step
token-PPO	0.761	150
turn-PPO	**0.963**	150
GRPO	0.881	265
GiGPO	<u>0.955</u>	200
HCAPO	0.784	145
HCAPO (adapted)	0.791	145
pre-RL init	0.440	0

Table 2: **ALFWorld final evaluation:** unseen 134-game success at each method's best checkpoint. Best value in bold, second best underlined.

Turn-PPO (0.963) and GiGPO (0.955) are effectively tied at the top; GRPO trails by ∼8 points, HCAPO by ∼17–18, and token-PPO is last at 0.761 — 20 points below its turn-level twin, which differs *only* in GAE granularity. On a 50-turn horizon, token-level value estimation is the clear failure mode.

4 Cross-environment reading

- **Horizon determines the winner.** Turn-level PPO dominates the 50-turn ALFWorld but is mid-pack on 4-turn SearchQA; GRPO's trajectory-level baseline is strongest on the short horizon. The token-vs-turn PPO contrast is the cleanest evidence: same critic, same budget, only the credit granularity differs, and the gap is +0.202 success on ALFWorld versus *(pending)* on SearchQA.
- **GiGPO is competitive without a critic** in its home environment (ALFWorld, 0.955) — *its SearchQA cell completes this claim's second half.*
- **HCAPO underperforms flat GRPO in both environments** under matched budgets (0.410–0.440 vs 0.463; 0.784–0.791 vs 0.881), reproducing the 2026-08-03 analysis.

5 Reproducibility

`assets/collect_results.py` pulls every number in this report (wandb curves, full-set eval tables, ALFWorld campaign CSVs) into `results/`; `assets/build_figures.py` renders both figures; `build_pdf.sh` compiles this document. Checkpoints: `/mnt/hdfs/mlsys/users/xiaoxuan/agentic_rl_ca/checkpoints/` (SearchQA) and `/mnt/hdfs/mlsys/users/xiaoxuan/alfworld_prm_gigpo/checkpoints/` (ALFWorld). New-arm launch infrastructure: `~/xiaoxuan/ca_cmp_jobs/` (job specs, entrypoint, resubmit loop).